COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010 Data Structure and Algorithm	**SECTION:**	CPE212
DATE PERFORMED:	08/26/26	**DATE SUBMITTED:**	08/26/26
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ACTIVITY NO.Prelim
Laboratory Exam Set D [Set D]

1. PROCEDURE OUTPUT

What procedures did you perform?

ALGORITHM

Start

1.Make a global declaration of maxCap for the stacks
2.Declare the prototype function
3.Make a switch case where you can input the pop
4.Input function Declaration
5.Pushing to the stack how we verify if the stack is Full
Input void push (){ to check the error

6 . Make an if statement that compares the top and bottom of the stack.

7

8

elim Skills > C SYMMETRIC_LL.H > ...

```
#ifndef SYMMETRIC_LL.H
#define SYMMETRIC_H
#include <iostream>

//creat a class node

template<typename T>
class stuckArr{

    if(top == -1) return true;

    bool isFull(){
        if(top == size -1) return true;
        return false;
    }

    bool isEmpty(){
        if(top == -1)return true;
        return false;
    }

    void push(char value){
        if(isFull()) {

        }

        stackArr[++top] = value;
    }

    char pop(){
        if(isEmpty())
        std::cout << "Stack Overflow\n";
        return ' ';

    }

    stackArr[++top] = value;

}
```

Prelim Skills > main.cpp > main()

```cpp
#include<iostream>
#include<string>

#define maxCap 100
int stackAr[maxCap];
int top = -1;

void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
bool dislay_stack(std::string);



int main(){
    // main driver code
    int choice;

    while (true){
        std::cout << "============================================" << std::endl;
        std::cout << "Stack Operations\n";
        std::cout << "1. PUSH 2. POP 3. TOP 4. isEmpty 5. isFull" << std::endl;
        std::cout << "============================================" << std::endl;
        std::cin >> choice;

switch(choice){
    case 1: push();
    break;
    case 2: pop();
    break;
    case 3:Top ();
    break;
    case 4: std::cout << "Is the stack empty? " << isEmpty() << std::endl;
    break;
    case 5: std::cout << "Is the stack full? " << isFull() << std::endl;
    break;
    default: std::cout << "Invalid choice" << std::endl;
    break;
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

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Purpose of Use: _____

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